SECJ3563 COMPUTATIONAL INTELLIGENCE

Assignment 1

Name:	Yong Jing Wen
Matric number:	A23CS0202
Section:	01

1.0 INTRODUCTION

Classification is one of the most significant methods in data mining and machine learning. It can help to predict the category of data based on its characteristics. Different classification algorithms will generate different results based on the dataset. In this study, we are using three classification algorithms which are J48, Random Forest and Random Tree. These 3 classification algorithms are applied to the Iris dataset using Weka tool to evaluate and compare their performance. Besides, the key evaluation metrics such as accuracy, error rate and Kappa statistics are used to measure the performance of each model. The purpose of this analysis is to identify the most effective classification algorithm for the given dataset while also examining their efficiency and interpretability.

2.0 OBJECTIVE

- To execute three different classification algorithms using 10-fold cross-validation in Weka.
- To evaluate the performance of each classifier based on their classification accuracy and error rates.
- To compare the results and identify which classifiers have the highest performance.

3.0 DESCRIPTION OF DATASET

To evaluate the performance of each classification, we are using the Iris dataset. This dataset consists of 150 instances and four numeric attributes which are Sepal length, Sepal width, Petal length and petal width that represent the physical dimension of the flowers. The target variable is the type of irises which are divided equally into three classes namely Iris-setosa, Iris-versicolor and Iris-virginica. Each class contains 50 instances.

4.0 ANALYSIS AND RESULT

The analysis results were obtained through 10 fold cross validation of J48, Random Forest and Random tree by using the Iris dataset.

Classifier output

```
petalwidth > 0.6
|   petalwidth <= 1.7
|   |   petallength <= 4.9: Iris-versicolor (48.0/1.0)
|   |   petallength > 4.9
|   |   |   petalwidth <= 1.5: Iris-virginica (3.0)
|   |   |   petalwidth > 1.5: Iris-versicolor (3.0/1.0)
|   petalwidth > 1.7: Iris-virginica (46.0/1.0)

Number of Leaves  :     5

Size of the tree :      9


Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         144               96      %
Incorrectly Classified Instances         6                4      %
Kappa statistic                          0.94
Mean absolute error                      0.035
Root mean squared error                  0.1586
Relative absolute error                  7.8705 %
Root relative squared error             33.6353 %
Total Number of Instances              150

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.980    0.000    1.000      0.980    0.990      0.985    0.990     0.987     Iris-setosa
                 0.940    0.030    0.940      0.940    0.940      0.910    0.952     0.880     Iris-versicolor
                 0.960    0.030    0.941      0.960    0.950      0.925    0.961     0.905     Iris-virginica
Weighted Avg.    0.960    0.020    0.960      0.960    0.960      0.940    0.968     0.924

=== Confusion Matrix ===

  a  b  c   <-- classified as
 49  1  0 |  a = Iris-setosa
  0 47  3 |  b = Iris-versicolor
  0  2 48 |  c = Iris-virginica
```

Analysis of J48 algorithms

The overall performance of the J48 classifier performs very well with a 96% accuracy and only has 6 misclassifications. In terms of reliability, it is based on the Kappa statistic. The kappa statistic for J48 is 0.94 shows that the model's predictions are closely matched with the actual iris classes and only a very small number of misclassifications exist. This indicates that the J48 model performs very reliably. In terms of efficiency and interpretability, J48 only takes 0 seconds to build the model and the resulting decision tree is highly concise with a total size of only 9 nodes and 5 leaves. This simplicity enhances the model's interpretability because it has a small size of tree which allows the human analysts to understand how decisions are made at each node easily.

Classifier output

```
              petallength
              petalwidth
              class
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

RandomForest

Bagging with 100 iterations and base learner

weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1 -do-not-check-capabilities

Time taken to build model: 0.04 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         143               95.3333 %
Incorrectly Classified Instances         7                4.6667 %
Kappa statistic                          0.93
Mean absolute error                      0.0408
Root mean squared error                  0.1621
Relative absolute error                  9.19   %
Root relative squared error             34.3846 %
Total Number of Instances              150

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Iris-setosa
                 0.940    0.040    0.922      0.940    0.931      0.896    0.991     0.984     Iris-versicolor
                 0.920    0.030    0.939      0.920    0.929      0.895    0.991     0.982     Iris-virginica
Weighted Avg.    0.953    0.023    0.953      0.953    0.953      0.930    0.994     0.989

=== Confusion Matrix ===

  a  b  c   <-- classified as
 50  0  0 |  a = Iris-setosa
  0 47  3 |  b = Iris-versicolor
  0  4 46 |  c = Iris-virginica
```

Analysis of Random Forest algorithms

The overall performance of Random Forest performed very closely to J48 with an accuracy of 95.33% and only 7 misclassifications. The Kappa statistic of Random Forest is 0.93 further confirms that it has a high reliability of the model. This indicated that the predicted class is almost perfectly consistent with the actual class. Regarding efficiency and Interpretability, the Random Forest took slightly longer to build the model which is 0.04 second because it needed to build 100 decision trees by using thousands of additional mathematical operations. Therefore, Random forests are less efficient and less interpretable than J48.

Classifier output

```
|   |   petallength >= 4.95
|   |   |   petalwidth < 1.55 : Iris-virginica (3/0)
|   |   |   petalwidth >= 1.55
|   |   |   |   sepallength < 6.95 : Iris-versicolor (2/0)
|   |   |   |   sepallength >= 6.95 : Iris-virginica (1/0)
|   petalwidth >= 1.75
|   |   petallength < 4.85
|   |   |   sepallength < 5.95 : Iris-versicolor (1/0)
|   |   |   sepallength >= 5.95 : Iris-virginica (2/0)
|   |   petallength >= 4.85 : Iris-virginica (43/0)

Size of the tree : 17

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         138               92      %
Incorrectly Classified Instances        12                8      %
Kappa statistic                          0.88
Mean absolute error                      0.0533
Root mean squared error                  0.2309
Relative absolute error                 12      %
Root relative squared error             48.9898 %
Total Number of Instances              150

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Iris-setosa
                 0.860    0.050    0.896      0.860    0.878      0.819    0.905     0.817     Iris-versicolor
                 0.900    0.070    0.865      0.900    0.882      0.822    0.915     0.812     Iris-virginica
Weighted Avg.    0.920    0.040    0.920      0.920    0.920      0.880    0.940     0.876

=== Confusion Matrix ===

  a  b  c   <-- classified as
 50  0  0 |  a = Iris-setosa
  0 43  7 |  b = Iris-versicolor
  0  5 45 |  c = Iris-virginica
```

Analysis of Random Tree algorithms

The Radom Tree has the lowest performance among the three models with an accuracy of 92% and has a 12 misclassification. This shows that the model is less effective in classifying the Iris dataset correctly. The Kappa statistic of Random Tree is 0.88 which still indicates a strong level of agreement between predicted class and actual class. Besides, the Radom tree took 0 seconds which shows a high computational efficiency. The tree size of the Random tree is 17 which is larger than J48 tree size. This means that the model is more complex and may have captured unnecessary details from training data.

5.0 CONCLUSION

Based on the analysis of the Iris dataset, the overall performance of the J48 algorithm has the highest performance followed by Random Forest and Random Tree . J48 achieved the highest accuracy and Kappa statistics compared to Random Forest and Random Tree. Besides, the tree size of the J48 is small, which makes it highly interpretable and easy to understand as well as maintain high efficiency with negligible training time. On the other hand, Random Forest performs almost as well as J48 with only a slight decrease in accuracy and reliability. This is because Random Forest involves multiple decision trees that lead to less interpretability and require more computation time rather than a single and clear structure. In addition, Random Tree shows the lowest performance with the lowest accuracy and a larger tree size. Although it is computationally efficient , its high complexity and poor predictive performance make it less suitable compared to J48 and Random Forest.

6.0 REFERENCES

Amrehn, M., Mualla, F., Angelopoulou, E., Steidl, S., & Maier, A. (2018). The random Forest classifier in WEKA: Discussion and new developments for imbalanced data. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.1812.08102

How Random trees classification and regression algorithm works—ArcGIS Pro | Documentation. (n.d.). https://pro.arcgis.com/en/pro-app/latest/tool-reference/geoai/how-random-trees-classification-and-regression-works.htm

GeeksforGeeks. (2023, March 6). *Building a machine learning model using J48 classifier*. GeeksforGeeks. https://www.geeksforgeeks.org/machine-learning/building-a-machine-learning-model-using-j48-classifier/